
SKILLS

- **Full Stack:** TypeScript, JavaScript, Python, Go, SQL, React, Node.js, Express, REST APIs, GraphQL, WebSocket
- **Backend Systems:** microservices, API design, event-driven workflows, HTTP, AMQP, TCP, caching, job queues
- **Data Stores:** PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Snowflake, schema design, query tuning
- **Trading/Analytics:** market data streams, backtesting, risk controls, dashboards, quantitative signals, position sizing
- **Cloud/Infra:** AWS, Kubernetes, Docker, Terraform, CI/CD, Linux, GitHub Actions, secure deployment patterns
- **Engineering Quality:** testing strategy, code reviews, design patterns, maintainability, ADRs, runbooks, documentation
- **Reliability:** OpenTelemetry, ELK, CloudWatch, tracing, alerting, SLOs, production debugging, incident response

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Full Stack / Platform Nov 2022 – Present
 - **0-to-1 Full-Stack Platform Delivery:** Led production delivery of a patent-pending enterprise blueprint platform using Node.js, TypeScript, React, REST APIs, real-time data rendering, rules-based recommendations, auto-certification workflows, and secure service matching; wrote 99.6% of production code and 100% of core logic.
 - **Internal Tooling for Technical Teams:** Built systems used by engineers, architects, security, infrastructure, and product stakeholders to analyze requirements, compare implementation options, generate architecture artifacts, and accelerate compliant delivery.
 - **Scalable Data & Observability Platform:** Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas for tracing, alerting, SLOs, production debugging, and downstream analytics.
 - **Architecture & Code Quality Leadership:** Led 5 engineers through technical planning, design reviews, code reviews, ADRs, documentation, runbooks, testing expectations, and operational handoff across regulated production systems.
 - **Distributed Integration Patterns:** Designed reusable service integration, gateway, IAM, data-handling, and cloud deployment patterns across AWS, GCP, OCP, Kubernetes, Terraform, API gateways, and enterprise security controls.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Startup Product Build:** Joined as first engineer and built the platform from scratch: React/TypeScript frontend, Node.js/Express APIs, Hyperledger Fabric verification layer, AWS EKS deployment, audit logging, and secure data workflows; team scaled to 20+ engineers before acquisition.
 - **Insurance Workflow Automation:** Designed validation and trust-score workflows for real-time premium verification and fraud-resistant claims processing; integrated document classification, monitoring, and HIPAA-aligned audit trails, reducing manual review by 60%.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Modern Web Platform:** Migrated federal tax platform UI from Apache Struts to React/TypeScript and contributed to backend API layers serving millions of Canadians with WCAG accessibility, automated quality gates, and maintainable code standards.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Software, Algorithms & Cloud:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Automatic Trading Algorithm:** Sub-second latency personal project integrating live brokerage APIs and real-time market data streams; momentum/mean-reversion signals with automated risk management, position sizing, and performance monitoring.
- **Portfolio Risk Optimization Framework:** Python analytics system for portfolio allocation, backtesting, Sharpe analysis, Monte Carlo simulation, market data ingestion, and risk-adjusted signal evaluation.
- **Flight Price Optimization Agent:** DQN/DDQN reinforcement-learning agent in Python for dynamic pricing; modelled demand curves on historical flight data and improved simulated outcome by 18% over baseline.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020